Linear Algebra and Its Applications

Section 8.6: Curves and Surfaces

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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8.6 Curves and surfaces

A cubic Bézier curve has four controls and formula

$$B(t) = (1-t)^3 p_0 + 3t(1-t)^2 p_1 + 3t^2(1-t)p_2 + t^3 p_3, \qquad 0 \le t \le 1.$$

The coefficients are nonnegative and sum to one, so the curve lies in the convex hull of its controls. A uniform cubic B-spline segment is a *different* curve, with weights

$$S(t) = \tfrac{1}{6}\big[(1-t)^3 p_0 + (3t^3 - 6t^2 + 4)p_1 + (-3t^3 + 3t^2 + 3t + 1)p_2 + t^3 p_3\big].$$

We use this formula, identical to the Practice Problems' equations (14)–(15), for the B-spline calculations below. Exercise 2 prints an inconsistent coefficient; its solution explicitly diagnoses the discrepancy before correcting it. The numbered exercises here concern curve segments and subdivision; the section's surface discussion uses these same one-variable operations in both parameter directions.

Practice problem 8.6.P1: The initial endpoint of a B-spline segment

Substitute $t = 0$ into $S(t)$ to obtain

$$S(0) = \tfrac{1}{6}p_0 + \tfrac{2}{3}p_1 + \tfrac{1}{6}p_2.$$

The coefficients are nonnegative and sum to one, so this point is in $\operatorname{conv}\{p_0, p_1, p_2\}$. Under the stated affine-independence assumption its unique affine coordinates are $(1/6, 2/3, 1/6)$. They differ from $(1, 0, 0)$, the coordinates of p_0, so it is not p_0 under that assumption. Without independence the formula still holds, but a degenerate choice such as all controls equal can make the endpoint coincide with a control.

Practice problem 8.6.P2: The final endpoint

At $t = 1$ the coefficients give

$$S(1) = \tfrac{1}{6}p_1 + \tfrac{2}{3}p_2 + \tfrac{1}{6}p_3.$$

Hence it is in $\operatorname{conv}\{p_1, p_2, p_3\}$. When those points are affinely independent, its affine coordinates are $(1/6, 2/3, 1/6)$, not $(0, 0, 1)$, so it differs from p_3. This proves the claimed endpoint behavior with its required nondegeneracy condition.

Exercise 8.6.1: Translate every control by the same vector

Write the four Bézier weights as $b_j(t)$. Their sum is $((1-t)+t)^3 = 1$. Therefore

$$B(t) + b = \sum_{j=0}^{3} b_j(t)p_j + \left(\sum_{j=0}^{3} b_j(t)\right) b = \sum_{j=0}^{3} b_j(t)(p_j + b).$$

This is a cubic Bézier curve with translated controls $p_j + b$. Translation is affine rather than linear, and the sum-one identity is exactly what makes the same control transformation work.

Exercise 8.6.2: Diagnose the printed coefficient before proving the B-spline properties

The exercise prints $3t(1-t) - 3t + 4$ as the coefficient of p_1 before the factor $1/6$, whereas the preceding Practice Problem equation (15) has $3t(1-t)^2 - 3t + 4$. The missing square changes the mathematics. With the printed unsquared expression, the four weights sum to

$$1 + \tfrac{1}{2}t^2(1-t),$$

not one. At $t = 1/2$ they sum to $17/16$. For example, if all four controls are one fixed nonzero vector p, the printed curve gives $(17/16)p$, which is outside their singleton convex hull. Thus part (a) is false literally as printed. We now use the consistent B-spline formula (14)–(15), with that square restored, for this exercise and later references to it.

(a) Prove convex-hull containment for the intended formula.

The first and fourth weights are $(1-t)^3/6$ and $t^3/6$, hence nonnegative. For the second use $3t(1-t)^2 - 3t + 4 \geq 4 - 3t \geq 1$ on $[0,1]$. For the third use $3t^2(1-t) + 3t + 1 \geq 1$. Expanding the four numerators makes their sum 6, so their weights sum to one. Thus $S(t)$ is a convex combination of the controls.

(b) Translate using the same sum-one property.

If the four weights are $s_j(t)$, then $S(t) + b = \sum_j s_j(t)(p_j + b)$. This is a B-spline with controls translated by b. The literal unsquared formula would not have this affine-invariance property: its translated-control expression adds $[1 + \frac{1}{2}t^2(1-t)]b$, not b.

Exercise 8.6.3: First and second derivatives of a cubic Bézier curve

(a) Differentiate the polynomial weights and regroup by differences.

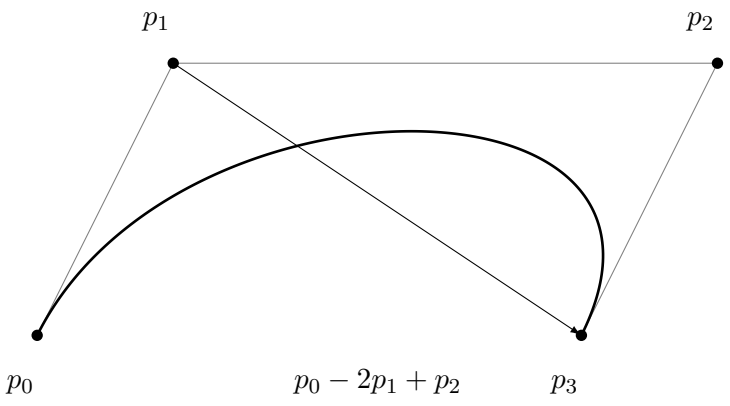
$$B'(t) = 3\big[(1-t)^2(p_1 - p_0) + 2t(1-t)(p_2 - p_1) + t^2(p_3 - p_2)\big].$$

In particular $B'(0) = 3(p_1 - p_0)$ and $B'(1) = 3(p_3 - p_2)$. The initial tangent direction goes from p_0 toward p_1; the final one goes from p_2 toward p_3, provided those differences are nonzero. Yes, $B'(1) = 0$ is possible: it occurs exactly when $p_3 = p_2$. At such an endpoint the zero tangent vector does not specify a direction.

(b) Differentiate once more.

$$B''(t) = 6\big[(1-t)(p_0 - 2p_1 + p_2) + t(p_1 - 2p_2 + p_3)\big].$$

Thus $B''(0) = 6(p_0 - 2p_1 + p_2)$ and $B''(1) = 6(p_1 - 2p_2 + p_3)$. With p_1 as origin, let $a = p_0 - p_1$, $c = p_2 - p_1$. Their vector sum $a + c = p_0 - 2p_1 + p_2$ has the direction of $B''(0)$. Construct the parallelogram on these two vectors; its diagonal gives the requested direction (unless the vector is zero). The schematic plot below uses controls $(0,0), (1,2), (5,2), (4,0)$ and shows that diagonal starting at p_1.



Exercise 8.6.4: Derivatives of the consistent cubic B-spline

(a) Compute the tangent and its endpoint values.

Differentiating the corrected formula gives

$$S'(t) = \tfrac{1}{2}\big[-(1-t)^2 p_0 + (3t^2 - 4t)p_1 + (-3t^2 + 2t + 1)p_2 + t^2 p_3\big].$$

Hence $S'(0) = (p_2 - p_0)/2$ and $S'(1) = (p_3 - p_1)/2$. The initial tangent is parallel to the segment from p_0 to p_2, and the final tangent to the segment from p_1 to p_3, when nonzero. If both vanish, then $p_2 = p_0 = a$ and $p_3 = p_1 = b$. Substitution gives

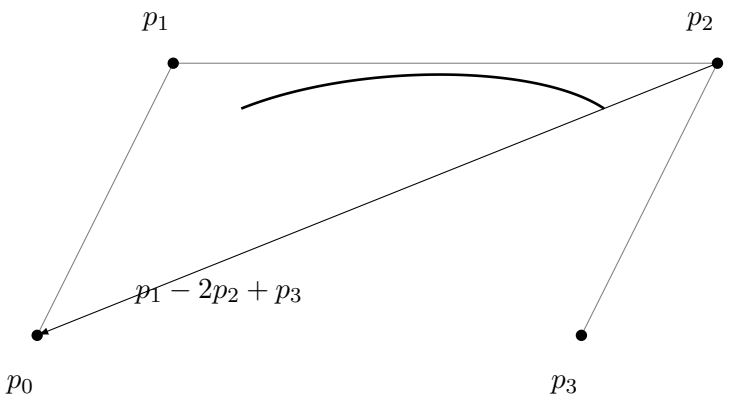
$$S(t) = (\tfrac{1}{3} + t^2 - \tfrac{2}{3}t^3)a + (\tfrac{2}{3} - t^2 + \tfrac{2}{3}t^3)b, \quad S'(t) = 2t(1-t)(a-b).$$

If $a \neq b$, this traverses the straight segment from $(a + 2b)/3$ to $(2a + b)/3$, with zero speeds at both endpoints but positive speed inside. It need not be constant. It is constant only if $a = b$.

(b) Compute the second derivative and its geometry.

$$S''(t) = (1-t)(p_0 - 2p_1 + p_2) + t(p_1 - 2p_2 + p_3).$$

In particular $S''(0) = p_0 - 2p_1 + p_2$ and $S''(1) = p_1 - 2p_2 + p_3$. Taking p_2 as origin, add the vectors $p_1 - p_2$ and $p_3 - p_2$; their parallelogram diagonal has the direction of $S''(1)$. The plot uses the same schematic controls as Exercise 8.6.3 but the B-spline curve and this second diagonal.



Exercise 8.6.5: Joining two cubic Bézier segments with matching velocities

(a) Equal derivatives give the midpoint condition.

At their shared control p_3, the first segment has derivative $3(p_3 - p_2)$ and the second has derivative $3(p_4 - p_3)$. Using the same unit parameter length for both pieces, C^1 continuity therefore requires and is guaranteed by

$$p_3 - p_2 = p_4 - p_3, \quad \text{or} \quad p_4 = 2p_3 - p_2.$$

Thus p_3 is the midpoint of p_2p_4.

(b) The zero-derivative case.

Both derivatives vanish exactly when $p_2 = p_3 = p_4$. The joined parametrization is still C^1, because its one-sided derivatives agree and are zero. But a zero velocity supplies no regular tangent direction; one cannot infer a smooth nonstationary join merely from parallel tangents. The remaining controls can give different departing and arriving directions at higher order, so C^2 continuity is not automatic.

Exercise 8.6.6: Adjacent B-spline segments automatically join

The first segment has controls p_0, p_1, p_2, p_3; the next has p_1, p_2, p_3, p_4.

(a) Check the endpoints.

By the endpoint formulas, $x(1) = (p_1 + 4p_2 + p_3)/6 = y(0)$, proving G^0 continuity.

(b) Check the derivatives with their actual scale factors.

The first segment has $x'(1) = (p_3 - p_1)/2$; the next has $y'(0) = (p_3 - p_1)/2$. They agree, so the combined unit-parameter curve is C^1 at the join. The last new control p_4 does not enter these equalities.

Exercise 8.6.7: A second derivative condition determines the next control

The C^1 condition from Exercise 8.6.5 gives $p_4 = 2p_3 - p_2$. The C^2 condition equates

$$x''(1) = 6(p_3 - 2p_2 + p_1), \qquad y''(0) = 6(p_3 - 2p_4 + p_5).$$

Cancel 6 and p_3, obtaining $p_5 = p_1 - 2p_2 + 2p_4$. Substitute the already determined p_4 to get

$$p_5 = p_1 - 4p_2 + 4p_3.$$

Thus p_4 and p_5 are fixed by the original controls and C^2 continuity; only the new segment's final control p_6 is free.

Exercise 8.6.8: Adjacent B-spline segments also match accelerations

For controls p_0, p_1, p_2, p_3, the final second derivative is $x''(1) = p_1 - 2p_2 + p_3$. For the next segment, with controls shifted forward by one, the initial second derivative is $y''(0) = p_1 - 2p_2 + p_3$. They are equal. Together with the endpoint and first-derivative equalities proved in Exercise 8.6.6, this establishes C^2 continuity at the join.

Exercise 8.6.9: The quartic Bézier basis matrix

Let $G = [\,p_0\ p_1\ p_2\ p_3\ p_4\,]$ and $u(t) = (1, t, t^2, t^3, t^4)^T$. Expand each of the five quartic weights in powers of t and put its coefficients in a *row* of M_B:

$$M_B = \begin{pmatrix} 1 & -4 & 6 & -4 & 1 \\ 0 & 4 & -12 & 12 & -4 \\ 0 & 0 & 6 & -12 & 6 \\ 0 & 0 & 0 & 4 & -4 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

For example its second row produces $4t - 12t^2 + 12t^3 - 4t^4 = 4t(1-t)^3$, the weight of p_1. Thus $x(t) = GM_Bu(t)$ reproduces the stated quartic curve.

Exercise 8.6.10: The uniform cubic B-spline basis matrix

Using $G = [\,p_0\ p_1\ p_2\ p_3\,]$ and $u(t) = (1, t, t^2, t^3)^T$, read the four consistent B-spline weights into rows:

$$M_S = \tfrac{1}{6}\begin{pmatrix} 1 & -3 & 3 & -1 \\ 4 & 0 & -6 & 3 \\ 1 & 3 & 3 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then $GM_Su(t)$ gives the B-spline formula from the Practice Problems. The factor $1/6$ is essential for normalization and cannot be dropped. The printed missing-square error in Exercise 2 is not used in this matrix.

Exercise 8.6.11: Three statements about Bézier curves

(a) **True.** A cubic Bézier curve uses four controls.

(b) **False.** For a quadratic curve the initial derivative is $2(p_1 - p_0)$, not $p_1 - p_0$. The direction is the same when nonzero, but the tangent vector has the extra factor two.

(c) **True.** The endpoint derivatives are $2(p_2 - p_1)$ and $2(p_3 - p_2)$, which agree exactly when $p_2 = (p_1 + p_3)/2$. The two pieces already share their endpoint, so this ensures C^1 continuity.

Exercise 8.6.12: Three statements about invariance and matrices

(a) **False.** Translations preserve the Bézier construction too, because the weights sum to one, as Exercise 8.6.1 proves.

(b) **True.** Equality of the ending and starting points is exactly G^0 continuity; matching derivatives requires an additional condition.

(c) **False.** The control columns form the geometry matrix G. The basis matrix contains fixed polynomial coefficients, independent of the controls.

Exercise 8.6.13: The first half's subdivided control points

(a) Use the endpoint derivative equation.

Equation (12) says $3(q_1 - q_0) = \frac{3}{2}(p_1 - p_0)$. Since $q_0 = p_0$, divide by 3 and rearrange: $q_1 = (p_0 + p_1)/2$.

(b) Rearrange the midpoint derivative equation.

Equation (13) gives $q_3 - q_2 = (-p_0 - p_1 + p_2 + p_3)/8$. Multiplying by eight and solving for q_2 gives $8q_2 = 8q_3 + p_0 + p_1 - p_2 - p_3$.

(c) Substitute the actual curve midpoint.

Use $q_3 = (p_0 + 3p_1 + 3p_2 + p_3)/8$ from equation (8). Then $8q_2 = 2p_0 + 4p_1 + 2p_2$, so

$$q_2 = (p_0 + 2p_1 + p_2)/4 = \tfrac{1}{2}[q_1 + (p_1 + p_2)/2].$$

Thus it is exactly the requested midpoint of two already constructed midpoints.

Exercise 8.6.14: The second half's subdivided controls

(a) Justify the endpoint derivative chain.

The equality $3(r_3 - r_2) = z'(1)$ is the Bézier endpoint formula. The equality $z'(1) = \frac{1}{2}x'(1)$ is the chain rule for $z(t) = x(1/2 + t/2)$. Finally $\frac{1}{2}x'(1) = \frac{3}{2}(p_3 - p_2)$ is the original endpoint formula.

(b) Solve for r_2.

Since $r_3 = p_3$, the equalities give $r_2 = (p_2 + p_3)/2$.

(c) Justify the initial derivative chain.

The identity $3(r_1 - r_0) = z'(0)$ is the initial Bézier derivative formula; $z'(0) = \frac{1}{2}x'(1/2)$ follows from the same reparametrization and chain rule.

(d) Use the original midpoint derivative.

Since $x'(1/2) = \frac{3}{4}(-p_0 - p_1 + p_2 + p_3)$, $3(r_1 - r_0) = \frac{3}{8}(-p_0 - p_1 + p_2 + p_3)$. Divide by 3 and rearrange to obtain $8r_1 = -p_0 - p_1 + p_2 + p_3 + 8r_0$.

(e) Substitute $r_0 = x(1/2)$.

The midpoint formula yields $8r_1 = 2p_1 + 4p_2 + 2p_3$. Consequently

$$r_1 = (p_1 + 2p_2 + p_3)/4 = \tfrac{1}{2}\big[r_2 + (p_1 + p_2)/2\big].$$

Exercise 8.6.15: A subdivision algorithm using only midpoint operations

(a) Apply the chain rule at the common midpoint.

With $y(t) = x(t/2)$ and $z(t) = x(1/2 + t/2)$, $y'(1) = \frac{1}{2}x'(1/2) = z'(0)$.

(b) Translate that equality into a control-point midpoint.

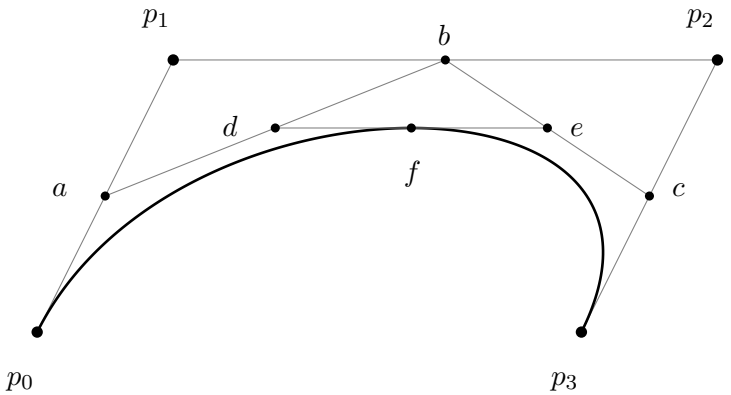
The endpoint formulas give $3(q_3 - q_2) = 3(r_1 - r_0)$. Using $q_3 = r_0$ and rearranging gives $2q_3 = q_2 + r_1$, so q_3 is the midpoint of $q_2 r_1$.

(c) State the full reusable algorithm.

Compute three first-level midpoints

$$a = (p_0 + p_1)/2, \quad b = (p_1 + p_2)/2, \quad c = (p_2 + p_3)/2;$$

then two second-level midpoints $d = (a + b)/2$, $e = (b + c)/2$; then the last midpoint $f = (d + e)/2$. The left controls are (p_0, a, d, f) and the right controls are (f, e, c, p_3). The formulas in Exercises 8.6.13–14 show $a = q_1, d = q_2, f = q_3 = r_0$, $e = r_1, c = r_2$. Thus the algorithm computes all required controls once, shares the repeated midpoint, and uses only sums and division by two.



Exercise 8.6.16: Endpoint positions and velocities determine the curve

The endpoint formulas recover all controls from the four specified vectors:

$$p_0 = x(0), \quad p_3 = x(1), \quad p_1 = x(0) + x'(0)/3, \quad p_2 = x(1) - x'(1)/3.$$

These equations determine one and only one control list, and substituting that list into the cubic Bézier formula determines one and only one curve. The derivative scale factor 3 is needed; just choosing any control points on the tangent lines would not determine the parameterized curve uniquely.

Exercise 8.6.17: Elevate a quadratic curve to cubic degree without changing it

(a) Match both endpoint positions and both endpoint velocities.

For $w(t) = (1-t)^2p_0+2t(1-t)p_1+t^2p_2$, $w(0) = p_0$, $w(1) = p_2$, $w'(0) = 2(p_1-p_0)$, and $w'(1) = 2(p_2-p_1)$. The preceding recovery formulas give cubic controls

$$r_0 = p_0, \quad r_1 = \tfrac{1}{3}p_0 + \tfrac{2}{3}p_1, \quad r_2 = \tfrac{2}{3}p_1 + \tfrac{1}{3}p_2, \quad r_3 = p_2.$$

Their endpoint derivatives are $3(r_1 - r_0) = w'(0)$ and $3(r_3 - r_2) = w'(1)$, so all four requirements are met.

(b) Check equality for every parameter.

Insert these controls into the cubic formula and collect the original p_0, p_1, p_2 coefficients. They become respectively

$$(1-t)^3 + t(1-t)^2 = (1-t)^2,$$
$$2t(1-t)^2 + 2t^2(1-t) = 2t(1-t), \qquad t^2(1-t) + t^3 = t^2.$$

Thus $x(t) = w(t)$ identically, including at both endpoints. The curve's polynomial degree representation has changed, but its geometry and parametrization have not.

Exercise 8.6.18: Multiply the scalar blocks by the control vectors

Treat the displayed p_j as vector blocks. Row-by-row block multiplication gives

$$\begin{pmatrix} p_0 \\ -3p_0 + 3p_1 \\ 3p_0 - 6p_1 + 3p_2 \\ -p_0 + 3p_1 - 3p_2 + p_3 \end{pmatrix}.$$

For controls in $\mathbb{R}^n$, this is a column of four vector blocks, or $4n$ scalar entries when expanded. These are the constant, linear, quadratic, and cubic vector coefficients of the curve; they agree with direct expansion of its four Bernstein weights.